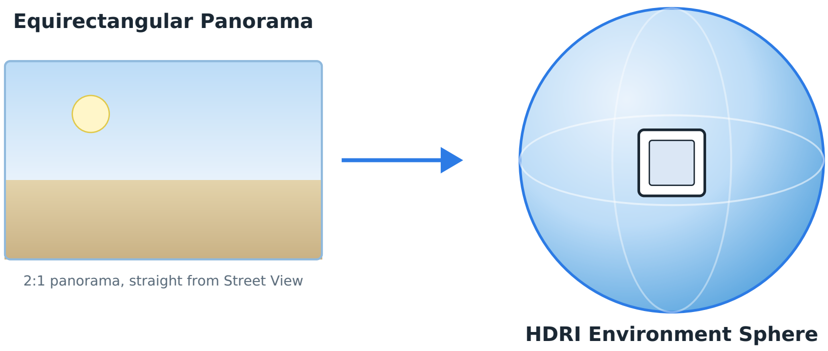
How to Create Your Own HDRI for Free Using Google Earth and Blender

Every 3D artist eventually hits the same wall: you find the perfect free HDRI online, but it’s not quite *your* location. Maybe you want the skyline from your hometown, a specific desert canyon, or a street corner that fits your scene’s mood exactly. The good news is that you don’t need a $2,000 360° camera or a panoramic tripod head to build your own environment map. With nothing but a browser, Google Maps, and Blender, you can turn almost any place on Earth into a usable HDRI in a few minutes.

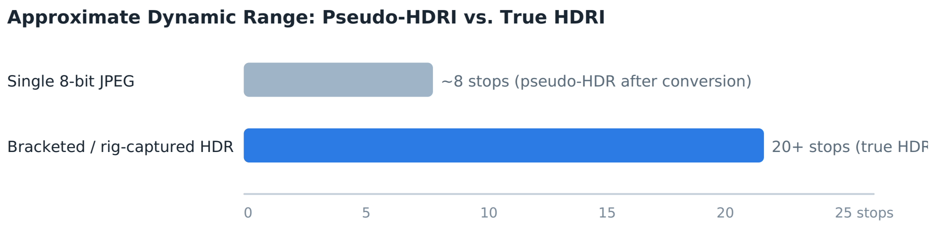


A Street View panorama gets wrapped around a sphere to light and reflect your whole 3D scene.

What Is an HDRI, Really?

HDRI stands for High Dynamic Range Image. Unlike a normal photo, which is stored with only 256 brightness levels per channel, a true HDRI can represent billions of distinct luminance values, which is why it can hold detail in both a blazing sky and a shadowed doorway at the same time. When you load an HDRI into Blender’s World shader, it’s not just a pretty backdrop: it becomes a giant sphere of light, so every reflective or diffuse surface in your scene picks up realistic color, contrast, and reflections from the environment. That’s why architectural visualization tools, product renders, and automotive showcases all lean so heavily on HDRI lighting instead of manually placing lamps.

It’s worth knowing upfront that there’s a real difference between a **true HDRI** and what the industry calls a **pseudo-HDRI**. A genuine HDRI is built by photographing the same scene at several exposures and merging them, or by using a specialized HDR panoramic rig, which lets the file capture 20 or more stops of dynamic range, far beyond the roughly 8 stops a single 8-bit JPEG can hold. The method in this guide, converting one JPEG panorama into an .hdr file, produces a pseudo-HDRI: it will look convincing as an environment and reflection map, but it can’t recover blown-out highlights or crushed shadows that were already lost when the JPEG was originally compressed, because wrapping a JPEG in an HDR container is a repackaging step, not a true reconstruction of lost dynamic range. For quick lookdev, mood boards, background plates, or hobby projects, this is more than good enough. For final production lighting on a hero shot, you’ll still want a properly bracketed HDRI or a high-quality free one from a library like Poly Haven.



A single converted JPEG carries far less dynamic range than a properly bracketed HDRI.

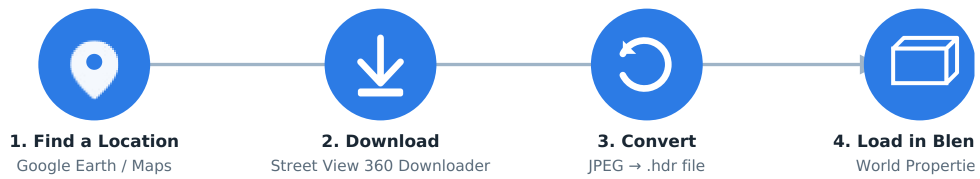
The Best Free HDRI Libraries (If You Just Need Something Fast)

Before building your own, it’s worth knowing where the 3D community already gets high-quality, properly captured HDRIs for free:

1. **ambientCG**: thousands of CC0 textures, HDRIs, and models with no attribution required.
2. **Poly Haven**: a widely trusted public library of real, bracketed HDRIs, textures, and 3D models.
3. **Textures.com**: a long-running resource for textures and photographic material, including panoramic skies.

These are excellent when you need something generic. But when the brief calls for *your* location, a specific city block, landmark, or landscape, building it yourself is the only option.

How to Build Your Own HDRI from Google Earth



The whole workflow: pick a spot, download it, convert it, then light your scene.

Step 1: Pick a Location in Google Earth or Google Maps

Open Google Earth or regular Google Maps and drop into Street View mode. Look for open, wide locations, such as deserts, plazas, coastlines, or hilltops, since these give the cleanest 360° coverage with fewer occlusions from nearby buildings or trees. Once you’ve found the spot, copy the URL from your browser’s address bar.

Step 2: Download the Panorama with Street View 360 Downloader

Instead of installing a desktop application, paste that URL directly into **Street View 360 Downloader**, a free browser-based tool that stitches and downloads the full equirectangular panorama for you. A few details that make it well suited for HDRI work specifically:

- **Resolution control**: choose from six zoom levels, from a quick 512×256 preview up to Maximum, which auto-fetches every available tile and can reach 8192×4096 or even 16384×8192 pixels depending on the location. Higher resolution matters a lot for HDRI, since low-res environment maps produce blurry reflections on glossy objects.
- **Multiple export formats**: save as JPEG, PNG, or WebP depending on what your HDR-conversion step needs.

- **Batch processing**: paste several URLs, Panorama IDs, or even mobile `maps.app.goo.gl` short links (one per line) and download an entire set of locations in one pass, handy if you're building a library of HDRIs for different times of day or seasons.
- **GPano XMP metadata option**: embeds standard 360° metadata into the file, so viewers, VR headsets, and platforms that expect equirectangular panoramas recognize the image correctly.
- No login, no API key, and files are generated on the fly rather than pulled from a stored library, so there's nothing to manage afterward.

Set the resolution as high as the location supports, pick JPEG (it plays nicest with most JPEG-to-HDR converters), and click download.

Step 3: Convert the JPEG Panorama into an .hdr File

Your downloaded panorama is still a standard JPEG at this point, and Blender's World shader wants a proper HDR/EXR container. Search for "JPEG to HDR converter" and you'll find several free online tools that wrap your image in the Radiance RGBE container Blender expects. This step doesn't invent new dynamic range: it repackages the JPEG's existing tonal data into a format 3D software will accept as an environment map, which is exactly why it's called a pseudo-HDR conversion rather than true HDR reconstruction. If your lighting setup later demands more precision, most of these converters also offer 32-bit OpenEXR output, which is the industry standard for VFX-grade lighting work.

Step 4: Load the HDRI into Blender

1. Switch to the **Rendered** viewport shading mode so you can preview the lighting live.
2. Open the **World Properties** tab and click the yellow dot next to *Color*.
3. Choose **Environment Texture**, then **Open**, and select the .hdr file you just converted.
4. Adjust the **Strength** slider in World Properties to control how bright the lighting from the panorama is: lower it for subtle fill light, raise it for a punchier, sun-drenched look.
5. If reflections look mirrored or rotated wrong, add a **Mapping node** between the Texture Coordinate and Environment Texture nodes and rotate the Z value until the horizon and sun position line up with your scene.

That's it: your scene is now lit and reflecting an environment map pulled directly from a real place on Earth.

Tips for Better Results

- **Shoot for locations with even, diffuse light.** Panoramas captured under harsh midday sun can create very hard-edged reflections once converted.
- **Watch the seam line.** Equirectangular panoramas can show a faint seam where the stitch wraps around; this is usually only visible if the camera is very close to reflective geometry.
- **Keep resolution proportional to your render distance.** A background HDRI visible only in reflections can get away with a lower resolution than one that's also the literal visible backdrop of the shot.
- **Batch a few exposures if you can.** If the same Street View location has been photographed on different dates or times of day, downloading a couple of versions gives you more lighting moods to pick from.

Final Thoughts

Turning a Street View panorama into a working Blender HDRI is a great shortcut when you need a specific, real-world location and don’t have professional HDR-capture gear on hand. It won’t fully replace a properly bracketed, production-grade HDRI for hero shots, but for concept art, background lighting, mood exploration, or personal projects, it’s a genuinely useful trick, and with a browser-based downloader instead of desktop software, the whole workflow from “interesting street corner” to “lit 3D scene” can take just a couple of minutes.